Positivity of the quartic solution of a discrete Painlevé-type recurrence

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Abstract. For $a > 0$, we prove the $d = 4$ case of a conjecture of Felder and Hoppe: every leading Gram matrix of their indefinite Hermitian form with density $\exp\bigl(-a|z|^2 + \frac{\mathrm{i}}{4}(z^4 + \bar{z}^4)\bigr)$ is nondegenerate, and all signed norms of the corresponding orthogonal polynomials are positive. Consequently, the solution of the quartic discrete Painlevé-type recurrence arising from the quantum minimal surface $w_2 = w_1^3$, with initial data given by Bessel functions of orders $\pm\frac{1}{4}$, is positive, and we show that it is the only positive solution of the recurrence. The proof writes the Bessel amplitude as a positive Laplace integral and uses independent source parameters for z^4 and $\bar{z}^4$ to express each of the four symmetry blocks through a bi-moment matrix with a positive weight. A triangular factorization and Andréief's identity then give every leading block minor as a prescribed sign times an explicit positive multiple integral. Uniqueness follows from a weighted energy estimate for the convex variational structure of the recurrence.

Keywords. discrete Painlevé-type recurrence; indefinite orthogonal polynomials; quartic potential; bi-moment determinant; quantum minimal surface.

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1. Introduction

Discrete Painlevé-type recurrences occur both in matrix regularizations of minimal surfaces and as recurrences for the coefficients of orthogonal polynomials. For the quantum minimal surfaces associated with the holomorphic curves $w_2 = w_1^{d-1}$, recurrences of this type were obtained in [CT99, AHK19, Hop25] (see [FH25, Section 3.1]). The quartic recurrence studied here is written explicitly in [Hop25, equation (38)] and [CDHM25, Section 5]; those works also treat the positive special-function solution in the cubic case $d = 3$.

Felder and Hoppe [FH25] constructed an indefinite Hermitian form for each d and showed that, whenever its leading Gram matrices are nondegenerate, the ratios of its orthogonal-polynomial norms obey

$$V_n + \frac{1}{a^2}\sum_{j=0}^{d-2}\prod_{k=0}^{d-2} V_{n+j-k} = \frac{n+1}{a}. \tag{1.1}$$

For $d = 4$, after setting $v_n = V_n/a$ and $\epsilon = a^{-2}$, the recurrence becomes

$$v_n\,(1 + v_{n-1}v_{n-2} + v_{n+1}v_{n-1} + v_{n+2}v_{n+1}) = (n+1)\epsilon, \qquad n \geq 0, \tag{1.2}$$

where $v_n = 0$ for $n < 0$. The two initial values are

$$v_0 = -\frac{1}{a}\frac{h'(a)}{h(a)}, \qquad v_1 = -\frac{1}{a}\frac{h''(a)}{h'(a)}, \tag{1.3}$$

with

$$\begin{aligned} h(a) = \frac{\pi^2 a}{4}\Bigg[& J_{-1/4}\left(\frac{a^2}{4}\right)^2 - \sqrt{2}\, J_{-1/4}\left(\frac{a^2}{4}\right) J_{1/4}\left(\frac{a^2}{4}\right) \\ & + J_{1/4}\left(\frac{a^2}{4}\right)^2 \Bigg]. \end{aligned} \tag{1.4}$$

This is the function $h(a) = \int_{\mathbb{C}} \rho_a \,\mathrm{d}A$ of [FH25, equation (3.4)] for $d = 4$ (the density ρ_a is recalled in (2.1)), in the Bessel form given in [FH25, Section 4]; Lemma 5.1 below rederives it. Felder and Hoppe proved the required Gram nondegeneracy for $d = 1, 2, 3$ and conjectured it for every d [FH25, Conjecture 3.6 and Theorem 3.7]; the quartic case is the first one not covered by their argument. Their conjecture is stated for $a \geq 0$. We treat $a > 0$, where all planar moments converge absolutely; at $a = 0$ they do not, and the endpoint would require an oscillatory or regularized interpretation, which we do not discuss.

Theorem 1.1. *Let $a > 0$.*

(i) *Every leading Gram matrix of the Felder–Hoppe Hermitian form for $d = 4$ (Section 2) is nondegenerate, and all its signed norms $\mathfrak{h}_n$ are positive. Thus [FH25, Conjecture 3.6] holds for $d = 4$ and $a > 0$.*

(ii) *The initial values (1.3)–(1.4) are well defined and extend uniquely to a real sequence satisfying (1.2) for every $n \geq 0$. This sequence is $v_n = \mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$; in particular, $v_n(a) > 0$ for every $n \geq 0$.*

Uniqueness in (ii) refers to the prescribed initial data. The next theorem shows that the positive solution itself is unique, so that the Bessel values (1.3) are the only initial values leading to a positive solution.

Theorem 1.2. *For every $\epsilon > 0$, there is at most one sequence of positive numbers $(v_n)_{n\geq 0}$ satisfying (1.2) for every $n \geq 0$. In particular, for $a > 0$ and $\epsilon = a^{-2}$, the sequence of Theorem 1.1 is the unique positive solution of (1.2).*

For the cubic analogue, the existence of a positive solution was proved in [AHK19, Section 7] and its uniqueness in [CDHM25, Theorem 2.1, proved in Section 4.2]; for the quartic recurrence, [CDHM25, Section 5] reports preliminary evidence for a positive solution with analogous properties. Our proof of Theorem 1.2 (Section 7) uses only the recurrence and is independent of the rest of the paper: in logarithmic variables the recurrence is the

stationarity condition of a convex functional, and a weighted energy estimate controls the boundary at infinity.

The proof is an integral sign mechanism that treats all indices at once. The square map $z \mapsto z^2$ and Gaussian integrals turn the quartic planar integral with coupling λ into a positive Laplace integral (Lemma 4.1). Independent source parameters u and v for z^4 and $\bar{z}^4$ are used on disks, where the two-source integral is entire and, by rotation invariance, depends only on uv. In exponential coordinates $u = \mathrm{e}^{t_1}$, $v = \mathrm{e}^{t_2}$ this turns the Gram block of each residue class modulo 4, after a unitriangular Stirling transformation, into the Hankel matrix of derivatives of a positive kernel $F_r(t)$ (Lemmas 4.2 and 4.3). Invertible triangular row and column transformations reduce these Hankel matrices to bi-moment matrices

$$\left[\int_0^\infty s^p \left(\frac{s^2}{1+s^2}\right)^q \mathrm{d}\mu_r(s)\right]_{p,q=0}^{k-1}$$

with positive measures μ_r. Since s and $s^2/(1+s^2)$ are both strictly increasing, Andréief's identity writes their determinants as positive integrals (Lemma 3.1). This fixes the sign of every leading block minor and gives an explicit positive multiple-integral formula for it (Proposition 4.4). The determinant step is a continuous Cauchy–Binet argument of the kind used for biorthogonal ensembles [Bor99]. To our knowledge, the new ingredient is the two-source representation: differentiation in a common quartic coupling mixes the insertions of z^4 and $\bar{z}^4$, whereas the (p, q) entry of a block requires their powers separately, and independent source derivatives select them. The construction uses two features of $d = 4$: the square map makes the quartic phase quadratic, and one integration by parts makes the kernels of the residue classes 2 and 3 manifestly positive. We do not consider higher d here.

Section 2 recalls the quartic Hermitian form. Section 3 proves the Hankel determinant identity. Section 4 derives the four positive kernels and evaluates the block determinants. Section 5 identifies the planar integral with the Bessel combination (1.4). Section 6 identifies the norm ratios with the solution of (1.2), and Section 7 proves Theorem 1.2. Theorems 1.1 and 1.2 have been formally verified in Lean 4.33.0 with Mathlib, as the declarations `Painleve.main_theorem` and `Painleve.unique_theorem`; the formalization follows the arguments given here, and the code availability statement at the end identifies the release.

2. The quartic Gram form

Fix $a > 0$ and put

$$\rho_a(z) = \exp\left(-a|z|^2 + \frac{\mathrm{i}}{4}(z^4 + \bar{z}^4)\right), \tag{2.1}$$

the density of [FH25, Section 3.1] for $d = 4$. The Gaussian factor makes all polynomial moments absolutely convergent. Let $\eta = \mathrm{e}^{\pi \mathrm{i}/4}$. Since $\eta^4 = -1$,

$$\rho_a(\mathrm{i}z) = \rho_a(z), \qquad \rho_a(\eta z) = \overline{\rho_a(z)}.$$

Decompose the polynomial ring into C_4-weight spaces

$$\mathbb{C}[z] = \bigoplus_{r=0}^{3} \mathcal{P}_r, \qquad \mathcal{P}_r = \{f \in \mathbb{C}[z] : f(\mathrm{i}z) = \mathrm{i}^r f(z)\} = \operatorname{span}\{z^{r+4q} : q \geq 0\}.$$

For $f \in \mathcal{P}_r$, set $(\sigma f)(z) = (-\eta)^{-r} f(\eta z)$, extend σ linearly to $\mathbb{C}[z]$, and define

$$(f,g)_a = \int_{\mathbb{C}} f(z)\overline{\sigma g(z)}\rho_a(z)\,\mathrm{d}A(z), \qquad \mathrm{d}A = \mathrm{d}(\mathrm{Re}\, z)\,\mathrm{d}(\mathrm{Im}\, z), \tag{2.2}$$

which is linear in f and conjugate-linear in g. This is the Hermitian form of [FH25, Proposition 3.1] for $d = 4$, with the right inverse $\chi(r) = r$ $(0 \le r \le 3)$ chosen there. If $m = r + 4q$ with $0 \le r \le 3$, then

$$\sigma z^m = (-\eta)^{-r}\eta^m z^m = (-1)^{r+q} z^m. \tag{2.3}$$

Since σ preserves each $\mathcal{P}_r$ and $\rho_a(\mathrm{i}z) = \rho_a(z)$, the substitution $z \mapsto \mathrm{i}z$ shows that distinct weight spaces are orthogonal. On $\mathcal{P}_r$, in the basis $z^r, z^{r+4}, \dots$, the Gram matrix is therefore

$$G^{(r)}_{pq} = (z^{r+4p}, z^{r+4q})_a = (-1)^{r+q} M^{(r)}_{pq}, \qquad M^{(r)}_{pq} = \int_{\mathbb{C}} z^{r+4p}\bar{z}^{r+4q}\rho_a(z)\,\mathrm{d}A(z). \tag{2.4}$$

The substitution $z \mapsto \eta z$ and $\rho_a(\eta z) = \overline{\rho_a(z)}$ give $\overline{M^{(r)}_{qp}} = (-1)^{q-p} M^{(r)}_{pq}$, so each $G^{(r)}$ is a Hermitian matrix; with the orthogonality of the weight spaces, this confirms directly that $(\ ,\ )_a$ is Hermitian.

For $k \ge 1$, let

$$\Delta^{(r)}_k = \det[G^{(r)}_{pq}]_{p,q=0}^{k-1}, \qquad \Delta^{(r)}_0 = 1. \tag{2.5}$$

Assume for the moment that all $\Delta^{(r)}_k$ are nonzero. After the monomials $1, z, \dots, z^N$ are grouped by residue modulo 4, their Gram matrix is block diagonal, with leading principal submatrices of the $G^{(r)}$ as blocks. Hence every leading Gram matrix $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ is nondegenerate, and Gram–Schmidt gives unique monic orthogonal polynomials $P_n(z) = z^n + \cdots$. Orthogonalizing z^n against lower monomials subtracts only components in the weight space of z^n, so $P_n \in \mathcal{P}_{n \bmod 4}$. Following [FH25, Lemma 3.4 and Example 3.5], for $\ell \in \mathbb{Z}$ put

$$\tau(\ell) = \begin{cases} -1, & \ell \equiv 3 \pmod 4, \\ 1, & \text{otherwise}, \end{cases} \qquad \varrho(n) = (-1)^n \prod_{\ell=0}^{n-1} \tau(\ell) = (-1)^{n+\lfloor n/4 \rfloor}. \tag{2.6}$$

The signed norms $\mathfrak{h}_n$ of [FH25, Section 3.1], denoted h_n there, are defined by

$$(P_n, P_n)_a = \varrho(n)\mathfrak{h}_n. \tag{2.7}$$

The matrices $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ are exactly the Gram matrices of [FH25, Conjecture 3.6]. Section 4 proves that all $\Delta^{(r)}_k$ are nonzero and that $\mathfrak{h}_n > 0$.

3. A Hankel determinant identity

Let $y > 0$, $\alpha \ge 0$, $\beta \ge 0$, and $b = \beta + \frac{1}{2}$. For $w > 0$, define

$$\Phi_{\alpha,\beta}(w; y) = \int_0^\infty \mathrm{e}^{-ys} s^{\alpha - 1/2}(1 + ws^2)^{-b}\,\mathrm{d}s. \tag{3.1}$$

Let μ be the measure $\mathrm{d}\mu(s) = \mathrm{e}^{-ys}s^{\alpha-1/2}(1+s^2)^{-b}\,\mathrm{d}s$ on $(0,\infty)$, put $\psi(s) = s^2/(1+s^2)$, and define the bi-moments

$$B_{pq}(y) = \int_0^\infty s^p \psi(s)^q \,\mathrm{d}\mu(s). \tag{3.2}$$

These integrals converge absolutely because $\alpha \geq 0$ and $y > 0$, and they may be differentiated under the integral sign, locally uniformly in $w, y > 0$: the derivative $\partial_w^j (1+ws^2)^{-b} = (-1)^j (b)_j s^{2j}(1+ws^2)^{-b-j}$ has absolute value at most $(b)_j w^{-j}$, and each y-derivative multiplies the integrand by $-s$. Here $(b)_j = b(b+1)\cdots(b+j-1)$ is the rising Pochhammer symbol. In particular, $\Phi_{\alpha,\beta}(\cdot\,;y)$ is smooth on $(0,\infty)$, and $\partial_y^i B_{0q} = (-1)^i B_{iq}$. We write $S_k = k(k-1)/2$ and $\left\{{q \atop j}\right\}$ for the Stirling numbers of the second kind, defined by $\left\{{0 \atop 0}\right\} = 1$, $\left\{{q \atop j}\right\} = 0$ for $j > q$ or $j = 0 < q$, and $\left\{{q+1 \atop j}\right\} = j\left\{{q \atop j}\right\} + \left\{{q \atop j-1}\right\}$.

Lemma 3.1 *(Hankel determinants of the kernel).* *Let $F(t) = \Phi_{\alpha,\beta}(\mathrm{e}^t; y)$. For every $k \geq 1$,*

$$(-1)^{S_k} \det\left[F^{(p+q)}(0)\right]_{p,q=0}^{k-1} = \left(\frac{y}{2}\right)^{S_k} \prod_{q=0}^{k-1} (b)_q \,\det[B_{pq}]_{p,q=0}^{k-1}, \tag{3.3}$$

and

$$\det[B_{pq}]_{p,q=0}^{k-1} = \frac{1}{k!}\int_{(0,\infty)^k} \prod_{1\leq i<j\leq k} (s_j - s_i)\big(\psi(s_j)-\psi(s_i)\big) \prod_{j=1}^{k} \mathrm{d}\mu(s_j) > 0. \tag{3.4}$$

In particular, $\operatorname{sgn}\det[F^{(p+q)}(0)]_{p,q=0}^{k-1} = (-1)^{S_k}$.

Proof. For a function G that is smooth on $(0,\infty)$, induction on q with the recurrence for $\left\{{q \atop j}\right\}$ gives

$$\frac{\mathrm{d}^q}{\mathrm{d}t^q} G(\mathrm{e}^t) = \sum_{j=0}^{q} \left\{{q \atop j}\right\} \mathrm{e}^{jt} G^{(j)}(\mathrm{e}^t), \tag{3.5}$$

because $\frac{\mathrm{d}}{\mathrm{d}t}\big[\mathrm{e}^{jt}G^{(j)}(\mathrm{e}^t)\big] = j\mathrm{e}^{jt}G^{(j)}(\mathrm{e}^t) + \mathrm{e}^{(j+1)t}G^{(j+1)}(\mathrm{e}^t)$. Differentiating (3.1) j times in w and substituting $s = x/\sqrt{w}$, which turns $(ws^2)^j(1+ws^2)^{-j}$ into $\psi(x)^j$, gives

$$w^j\,\partial_w^j \Phi_{\alpha,\beta}(w;y) = (-1)^j (b)_j\, w^{-\gamma} B_{0j}\big(yw^{-1/2}\big), \qquad \gamma = \frac{\alpha}{2} + \frac{1}{4}. \tag{3.6}$$

Hence, by (3.5) with $G = \Phi_{\alpha,\beta}(\cdot\,;y)$,

$$F^{(q)}(t) = \sum_{j=0}^{q} \left\{{q \atop j}\right\} (-1)^j (b)_j\, \mathrm{e}^{-\gamma t} B_{0j}\big(y\mathrm{e}^{-t/2}\big).$$

Next, for a function g that is smooth on $(0,\infty)$, put $Y = y\mathrm{e}^{-t/2}$; since $\frac{\mathrm{d}}{\mathrm{d}t}\big[\mathrm{e}^{-\gamma t}Y^i g^{(i)}(Y)\big] = \mathrm{e}^{-\gamma t}\big[(-\gamma - \tfrac{i}{2})Y^i g^{(i)}(Y) - \tfrac{1}{2}Y^{i+1}g^{(i+1)}(Y)\big]$, induction on p gives

$$\frac{\mathrm{d}^p}{\mathrm{d}t^p}\Big[\mathrm{e}^{-\gamma t} g\big(y\mathrm{e}^{-t/2}\big)\Big]\Big|_{t=0} = \sum_{i=0}^{p} \kappa_{pi}\, y^i g^{(i)}(y)$$

with constants κ_{pi} depending only on γ, determined by $\kappa_{00} = 1$ and $\kappa_{p+1,i} = (-\gamma - \frac{i}{2})\kappa_{pi} - \frac{1}{2}\kappa_{p,i-1}$, with $\kappa_{pi} = 0$ for $i < 0$ or $i > p$; in particular $\kappa_{pp} = (-\frac{1}{2})^p$. Applying this to $g = B_{0j}$, with $g^{(i)} = (-1)^i B_{ij}$, we obtain

$$F^{(p+q)}(0) \;=\; \sum_{i=0}^{p}\sum_{j=0}^{q} \kappa_{pi}(-y)^i\, B_{ij}\,(-1)^j (b)_j \left\{ {q \atop j} \right\}.$$

Thus the $k \times k$ truncations satisfy $[F^{(p+q)}(0)] = TB\Lambda\Sigma^{\mathsf{T}}$, where $T_{pi} = \kappa_{pi}(-y)^i$ is lower triangular with $T_{pp} = (y/2)^p$, $\Lambda = \operatorname{diag}\big((-1)^j(b)_j\big)_{j=0}^{k-1}$, and $\Sigma_{qj} = \left\{ {q \atop j} \right\}$ is lower triangular with ones on the diagonal. Hence $\det T = (y/2)^{S_k}$, $\det \Sigma = 1$, and $\det \Lambda = (-1)^{S_k} \prod_{j=0}^{k-1}(b)_j$. This proves (3.3).

Andréief's identity [And86, For19], applied to the functions s^p and $\psi(s)^q$, all of whose products are μ-integrable, gives

$$\det[B_{pq}]_{p,q=0}^{k-1} \;=\; \frac{1}{k!}\int_{(0,\infty)^k} \det[s_j^p]_{p=0,\dots,k-1}^{j=1,\dots,k}\, \det[\psi(s_j)^q]_{q=0,\dots,k-1}^{j=1,\dots,k} \prod_{j=1}^{k} \mathrm{d}\mu(s_j).$$

Both determinants are Vandermonde determinants, which gives the integrand in (3.4). Since s and ψ are strictly increasing on $(0,\infty)$, this integrand is positive off the diagonals $s_i = s_j$, which are $\mu^{\otimes k}$-null, and μ has a positive density. Hence $\det B > 0$. Finally, $(b)_q > 0$ because $b > 0$. ∎

4. The four positive kernels

Throughout this section,

$$y \;=\; \frac{a^2}{2}, \qquad C \;=\; \sqrt{\frac{\pi}{2}}.$$

Lemma 4.1 *(positive source).* *For every $\lambda > 0$,*

$$\int_{\mathbb{C}} \exp\left(-a|z|^2 + \frac{\mathrm{i}\lambda}{4}(z^4 + \bar{z}^4)\right) \mathrm{d}A(z) \;=\; C\,\Phi_{0,0}(\lambda^2; y). \tag{4.1}$$

Proof. The integrand is bounded by $\mathrm{e}^{-a|z|^2}$. The map $z \mapsto w = z^2$ is two-to-one from $\mathbb{C}\setminus\{0\}$ onto itself with Jacobian $4|z|^2 = 4|w|$, and $z^4 + \bar{z}^4 = 2(X^2 - Y^2)$ for $w = X + \mathrm{i}Y$. Hence the left-hand side of (4.1) equals

$$\frac{1}{2}\int_{\mathbb{R}^2} \frac{\mathrm{e}^{-a|w|}}{|w|}\, \mathrm{e}^{\mathrm{i}\lambda(X^2-Y^2)/2}\, \mathrm{d}X\, \mathrm{d}Y.$$

By [OLBC10, equations (10.32.10) and (10.39.2)] with $\nu = -\frac{1}{2}$,

$$\frac{\mathrm{e}^{-a|w|}}{|w|} \;=\; \frac{1}{\sqrt{\pi}}\int_0^\infty \sigma^{-1/2} \exp\left(-\sigma|w|^2 - \frac{a^2}{4\sigma}\right) \mathrm{d}\sigma.$$

The resulting triple integral converges absolutely, because $\int_{\mathbb{R}^2} \mathrm{e}^{-\sigma|w|^2}\,\mathrm{d}X\,\mathrm{d}Y = \pi/\sigma$ and $\int_0^\infty \sigma^{-3/2}\mathrm{e}^{-a^2/(4\sigma)}\,\mathrm{d}\sigma < \infty$. By Fubini's theorem and the Gaussian integrals

$\int_{\mathbb{R}} \mathrm{e}^{-(\sigma \mp \mathrm{i}\lambda/2)X^2}\,\mathrm{d}X = \sqrt{\pi}\,(\sigma \mp \mathrm{i}\lambda/2)^{-1/2}$ (principal branch), whose product is $\pi(\sigma^2 + \lambda^2/4)^{-1/2}$, the left-hand side equals

$$\frac{\sqrt{\pi}}{2}\int_0^\infty \sigma^{-1/2}\mathrm{e}^{-a^2/(4\sigma)}\left(\sigma^2+\frac{\lambda^2}{4}\right)^{-1/2}\mathrm{d}\sigma.$$

The substitution $\sigma = 1/(2s)$, for which $\sigma^2 + \lambda^2/4 = (1+\lambda^2 s^2)/(4s^2)$, turns this into

$$C\int_0^\infty \frac{\mathrm{e}^{-ys}}{\sqrt{s(1+\lambda^2 s^2)}}\,\mathrm{d}s = C\,\Phi_{0,0}(\lambda^2; y). \qquad \blacksquare$$

For $r \in \{0,1,2,3\}$ and $\lambda > 0$, define the weighted sources

$$\mathcal{W}_r(a;\lambda) = \int_{\mathbb{C}} |z|^{2r}\exp\left(-a|z|^2 + \frac{\mathrm{i}\lambda}{4}(z^4+\bar z^4)\right)\mathrm{d}A(z).$$

Differentiation under the integral sign, dominated on $a \ge a_0 > 0$ by $|z|^{2r+2}\mathrm{e}^{-a_0|z|^2}$, gives $\mathcal{W}_{r+1} = -\partial_a \mathcal{W}_r$ for $r = 0,1,2$, so $\mathcal{W}_r = (-\partial_a)^r \mathcal{W}_0$.

Lemma 4.2 *(four positive kernels).* *Define*

r	A_r	α_r	β_r
0	C	0	0
1	aC	1	0
2	$2C$	1	1
3	$2aC$	2	1

(4.2)

Then $\mathcal{W}_r(a;\lambda) = A_r\Phi_{\alpha_r,\beta_r}(\lambda^2; y)$ for $r = 0,1,2,3$ and $\lambda > 0$.

Proof. Write $w = \lambda^2$. The case $r = 0$ is Lemma 4.1. Since $\partial_a = a\partial_y$ and $\partial_y\Phi_{\alpha,\beta} = -\Phi_{\alpha+1,\beta}$, the case $r = 1$ follows. Next, $(-\partial_a)^2 = \partial_y + 2y\partial_y^2$, so $\mathcal{W}_2 = C\big(2y\Phi_{2,0} - \Phi_{1,0}\big)(w;y)$. Integration by parts, whose boundary terms vanish at 0 and ∞, gives

$$y\int_0^\infty \frac{\mathrm{e}^{-ys}s^{3/2}}{\sqrt{1+ws^2}}\,\mathrm{d}s = \frac12\int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{\sqrt{1+ws^2}}\,\mathrm{d}s + \int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{(1+ws^2)^{3/2}}\,\mathrm{d}s, \tag{4.3}$$

that is, $y\Phi_{2,0} = \frac12\Phi_{1,0} + \Phi_{1,1}$. Hence $\mathcal{W}_2 = 2C\Phi_{1,1}$, and $\mathcal{W}_3 = -a\partial_y\mathcal{W}_2 = 2aC\Phi_{2,1}$. $\blacksquare$

Lemma 4.3 *(the Gram blocks in exponential coordinates).* *For $r \in \{0,1,2,3\}$ let $M^{(r)}_{kl} = \int_{\mathbb{C}} z^{r+4k}\bar z^{r+4l}\rho_a(z)\,\mathrm{d}A(z)$ and $F_r(t) = A_r\Phi_{\alpha_r,\beta_r}(\mathrm{e}^t; y)$. Then, for all $p,q \ge 0$,*

$$\sum_{k=0}^{p}\sum_{l=0}^{q}\left\{{p \atop k}\right\}\left\{{q \atop l}\right\}\left(\frac{\mathrm{i}}{4}\right)^{k+l} M^{(r)}_{kl} = F_r^{(p+q)}(0). \tag{4.4}$$

Proof. For $R > 0$ and $u, v \in \mathbb{C}$, define the disk integral

$$\mathcal{Z}_R(u,v) = \int_{|z|\le R} |z|^{2r}\exp\left(-a|z|^2 + \frac{\mathrm{i}}{4}(uz^4 + v\bar z^4)\right)\mathrm{d}A(z).$$

Over the compact disk, all derivatives in (u,v) may be taken under the integral sign, and $\mathcal{Z}_R$ is entire in each variable.

Step 1: $\mathcal{Z}_R(u,v)=\mathcal{Z}_R(uv,1)$ for $u,v>0$. For fixed $u,v>0$, the function $H(s)=\mathcal{Z}_R(ue^{s},ve^{-s})$ is entire in $s\in\mathbb{C}$. The rotation $z\mapsto e^{i\varphi/4}z$, $\varphi\in\mathbb{R}$, preserves the disk, $|z|$ and the area measure, and multiplies z^4 by $e^{i\varphi}$ and $\bar z^4$ by $e^{-i\varphi}$; hence $H(s+i\varphi)=H(s)$. Differentiating in φ at $\varphi=0$ gives $iH'(s)=0$, so H is constant, and $H(0)=H(\log v)$ is the claim.

Step 2: the truncated identity. For real t_1,t_2 put $K(t_1,t_2)=\mathcal{Z}_R(e^{t_1},e^{t_2})$. By Step 1, $K(t_1,t_2)=f_R(t_1+t_2)$ with $f_R(t)=\mathcal{Z}_R(e^{t/2},e^{t/2})$, so $\partial_{t_2}^q\partial_{t_1}^pK(0,0)=f_R^{(p+q)}(0)$. On the other hand, (3.5) with $G(w)=\exp(cw)$ gives $\frac{d^p}{dt^p}\exp(ce^t)=\sum_k\left\{{p\atop k}\right\}(ce^t)^k\exp(ce^t)$; applying this under the integral, first with $c=iz^4/4$ and then with $c=i\bar z^4/4$, yields

$$\partial_{t_2}^q\partial_{t_1}^pK(0,0)=\sum_{k=0}^{p}\sum_{l=0}^{q}\left\{{p\atop k}\right\}\left\{{q\atop l}\right\}\left(\frac{i}{4}\right)^{k+l}\int_{|z|\le R}z^{r+4k}\bar z^{r+4l}\rho_a(z)\,dA(z),$$

where $|z|^{2r}z^{4k}\bar z^{4l}=z^{r+4k}\bar z^{r+4l}$.

Step 3: $R\to\infty$. The truncated moments converge to $M_{kl}^{(r)}$, since their integrands are bounded by $|z|^{2r+4k+4l}e^{-a|z|^2}$. For real t, the factor $\exp\bigl(\frac{i}{4}e^{t/2}(z^4+\bar z^4)\bigr)$ has modulus 1, and by (3.5) its t-derivatives of order m are this factor times a polynomial in $e^{t/2}(z^4+\bar z^4)$; for $|t|\le1$ they are bounded by $c_m(1+|z|)^{4m}$. Hence derivatives of all orders may be taken under the integral both for f_R and for its full-plane counterpart $t\mapsto\mathcal{W}_r(a;e^{t/2})$, and dominated convergence gives $f_R^{(m)}(0)\to\frac{d^m}{dt^m}\mathcal{W}_r(a;e^{t/2})|_{t=0}$. By Lemma 4.2 with $\lambda=e^{t/2}$, $\mathcal{W}_r(a;e^{t/2})=F_r(t)$. Letting $R\to\infty$ in Step 2 proves (4.4). ∎

Proposition 4.4 *(block determinants).* *For $r\in\{0,1,2,3\}$ and $k\ge1$, with the parameters (4.2),*

$$\begin{aligned}(-1)^{rk+S_k}\Delta_k^{(r)}&=\frac{A_r^k(4a^2)^{S_k}}{k!}\prod_{q=0}^{k-1}\Bigl(\beta_r+\frac12\Bigr)_q\\&\quad\times\int_{(0,\infty)^k}\prod_{j=1}^{k}\frac{e^{-ys_j}s_j^{\alpha_r-1/2}}{(1+s_j^2)^{\beta_r+k-1/2}}\prod_{1\le i<j\le k}(s_j-s_i)^2(s_i+s_j)\prod_{j=1}^{k}ds_j.\end{aligned}\tag{4.5}$$

The right-hand side is strictly positive, so $\operatorname{sgn}\Delta_k^{(r)}=(-1)^{rk+S_k}$.

Proof. By (2.4), $G_{pq}^{(r)}=(-1)^{r+q}(4/i)^{p+q}\bigl[(i/4)^{p+q}M_{pq}^{(r)}\bigr]$. Taking $(4/i)^p$ out of row p and $(-1)^{r+q}(4/i)^q$ out of column q, and using $(4/i)^2=-16$, gives

$$\Delta_k^{(r)}=(-1)^{rk}16^{S_k}\det\left[\left(\frac{i}{4}\right)^{p+q}M_{pq}^{(r)}\right]_{p,q=0}^{k-1}.$$

The left-hand side of (4.4) is the (p,q) entry of $\Sigma\,[(i/4)^{k+l}M_{kl}^{(r)}]\,\Sigma^{\mathsf T}$, with Σ the unitriangular Stirling matrix of Lemma 3.1; its $k\times k$ truncations therefore have the same determinant. By Lemma 4.3 that determinant equals $\det[F_r^{(p+q)}(0)]=A_r^k\det[F^{(p+q)}(0)]$, where F is the

function of Lemma 3.1 for $(\alpha,\beta) = (\alpha_r,\beta_r)$. Hence that lemma and $16^{S_k}(y/2)^{S_k} = (4a^2)^{S_k}$ give

$$(-1)^{rk+S_k}\Delta_k^{(r)} = A_r^k(4a^2)^{S_k}\prod_{q=0}^{k-1}\left(\beta_r+\frac{1}{2}\right)_q \det[B_{pq}]_{p,q=0}^{k-1}.$$

Finally,

$$\psi(s_j)-\psi(s_i) = \frac{(s_j-s_i)(s_j+s_i)}{(1+s_i^2)(1+s_j^2)},$$

and each factor $1+s_j^2$ occurs $k-1$ times in the product over $i<j$. Substituting this into (3.4) gives (4.5), and its positivity is part of Lemma 3.1. ∎

Corollary 4.5 *(quartic Gram nondegeneracy).* *For every $N \geq 0$, the leading Gram matrix $\big[(z^p,z^q)_a\big]_{p,q=0}^{N}$ is nondegenerate, and the signed norms in (2.7) satisfy*

$$\mathfrak{h}_n > 0 \qquad (n \geq 0).$$

In particular, [FH25, Conjecture 3.6] holds for $d=4$ and $a>0$.

Proof. By Proposition 4.4, every $\Delta_k^{(r)}$ is nonzero, so by Section 2 every leading Gram matrix is nondegenerate and the polynomials P_n exist. Let $n = r+4j$. Replacing z^n by P_n in the basis $z^r, z^{r+4}, \dots, z^{r+4j}$ is a unitriangular change of basis, and orthogonality makes the new Gram matrix $\operatorname{diag}\big([G_{pq}^{(r)}]_{p,q=0}^{j-1}, (P_n,P_n)_a\big)$. Comparing determinants gives

$$(P_n,P_n)_a = \frac{\Delta_{j+1}^{(r)}}{\Delta_j^{(r)}}; \tag{4.6}$$

this Schur-complement identity uses only the nonvanishing of the leading minors, not positive definiteness. By Proposition 4.4, $\operatorname{sgn}(P_n,P_n)_a = (-1)^{r+j}$, and (2.6) gives $\varrho(r+4j) = (-1)^{r+j}$. Hence $\mathfrak{h}_n = \varrho(n)(P_n,P_n)_a > 0$. The last assertion is the identification at the end of Section 2. ∎

Remark 4.6. By (4.6), $\mathfrak{h}_n = |\Delta_{j+1}^{(r)}|/|\Delta_j^{(r)}|$ for $n = r+4j$, and (4.5) writes each determinant as an integral of a positive function. The norms, and the ratios $v_n = \mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$ of Section 6, can therefore be evaluated without the cancellation inherent in the oscillatory moment determinants.

5. The Bessel form of the initial function

Lemma 5.1 *(Bessel form).* *For every $a>0$, $\int_{\mathbb{C}}\rho_a\,\mathrm{d}A = h(a)$, the Bessel combination (1.4).*

Proof. Put $H(a) = \int_{\mathbb{C}}\rho_a\,\mathrm{d}A$ and $m_{pq} = \int_{\mathbb{C}} z^p\bar z^q\rho_a(z)\,\mathrm{d}A(z)$ for $a>0$.

Step 1: $H'''+a^2H'+aH = 0$ on $(0,\infty)$. Differentiation under the integral sign gives $H^{(k)}(a) = (-1)^k m_{kk}$. With the Wirtinger derivative $\partial_z = \frac{1}{2}(\partial_{\operatorname{Re} z} - \mathrm{i}\partial_{\operatorname{Im} z})$, $\partial_z\bar z^q = 0$ and

$\partial_z \rho_a = (-a\bar z + \mathrm{i}z^3)\rho_a$, so integrating $\partial_z(z^p \bar z^q \rho_a)$ over $\mathbb{C}$, with no boundary terms because of the Gaussian decay, gives

$$p\, m_{p-1,q} + \mathrm{i}\, m_{p+3,q} \;=\; a\, m_{p,q+1}.$$

For $(p,q) = (1,0)$ and $(0,3)$ this reads $m_{00} + \mathrm{i}m_{40} = a m_{11}$ and $\mathrm{i}m_{33} = a m_{04}$. The substitution $z \mapsto \eta z$ of Section 2 gives $\overline{m_{qp}} = \bar\eta^{\,p}\eta^q m_{pq}$; hence every m_{kk} is real and $\overline{m_{40}} = -m_{04}$. Therefore $m_{33} = a m_{00} - a^2 m_{11}$, which is the claimed equation.

Step 2: the Bessel side. For $x > 0$, $J_\nu(x) = (x/2)^\nu s_\nu(x)$ with the even entire function $s_\nu(x) = \sum_{k\ge 0} (-1)^k x^{2k} / \big(4^k k!\,\Gamma(k+\nu+1)\big)$ [OLBC10, equation (10.2.2)], and Bessel's equation [OLBC10, equation (10.2.1)] becomes $x s_\nu'' + (2\nu+1) s_\nu' + x s_\nu = 0$. Writing $s_\pm = s_{\pm 1/4}$ and $x = a^2/4$, so that $(x/2)^{\mp 1/4} = (a^2/8)^{\mp 1/4}$, we get, for $a > 0$,

$$h(a) \;=\; h_S(a) \;:=\; \frac{\pi^2}{4}\left[\sqrt{8}\, s_-(x)^2 - \sqrt{2}\, a\, s_-(x) s_+(x) + \frac{a^2}{\sqrt{8}}\, s_+(x)^2\right].$$

The function h_S is smooth on $\mathbb{R}$. Put $U(a) = s_-(a^2/4)$ and $V(a) = a\, s_+(a^2/4)$. The equations for $s_\mp$, with $2\nu+1 = \frac12$ and $\frac32$, give $U'' + \frac{a^2}{4} U = 0$ and $V'' + \frac{a^2}{4} V = 0$. If y_1, y_2 solve $y'' + qy = 0$, then $P = y_1 y_2$ satisfies $P''' + 4qP' + 2q'P = 0$; with $q = a^2/4$ this is $P''' + a^2 P' + aP = 0$. Since $h_S = \frac{\pi^2}{4}\big(\sqrt{8}\, U^2 - \sqrt{2}\, UV + V^2/\sqrt{8}\big)$, we get $h_S''' + a^2 h_S' + a h_S = 0$ on $\mathbb{R}$. Since $s_\nu(0) = 1/\Gamma(\nu+1)$ and $s_\nu'(0) = 0$,

$$h_S(0) \;=\; \frac{\pi^2\sqrt{8}}{4\,\Gamma(\frac34)^2}, \qquad h_S'(0) \;=\; -\frac{\pi^2\sqrt{2}}{4\,\Gamma(\frac34)\Gamma(\frac54)}, \qquad h_S''(0) \;=\; \frac{\pi^2}{2\sqrt{8}\,\Gamma(\frac54)^2}.$$

Step 3: the planar side as $a \to 0^+$. By Lemmas 4.1 and 4.2 with $\lambda = 1$,

$$\begin{aligned} H &= C\,\Phi_{0,0}(1;y), \\ H' &= -\mathcal{W}_1(a;1) = -aC\,\Phi_{1,0}(1;y), \\ H'' &= \mathcal{W}_2(a;1) = 2C\,\Phi_{1,1}(1;y). \end{aligned}$$

As $y \downarrow 0$, monotone convergence and $\int_0^\infty s^{2x-1}(1+s^2)^{-x-z}\,\mathrm{d}s = \frac12 B(x,z) = \frac{\Gamma(x)\Gamma(z)}{2\Gamma(x+z)}$ [OLBC10, equations (5.12.1) and (5.12.3)] give $\Phi_{0,0}(1;y) \to \frac12 B(\frac14,\frac14) = \Gamma(\frac14)^2/(2\sqrt{\pi})$ and $\Phi_{1,1}(1;y) \to \frac12 B(\frac34,\frac34) = \Gamma(\frac34)^2/\sqrt{\pi}$. The substitution $s = u/y$ gives $a\Phi_{1,0}(1;y) = \sqrt{2}\int_0^\infty \mathrm{e}^{-u} u^{1/2}(y^2+u^2)^{-1/2}\,\mathrm{d}u$; the integrand is bounded by $\mathrm{e}^{-u}u^{-1/2}$, so by dominated convergence this tends to $\sqrt{2}\,\Gamma(\frac12) = \sqrt{2\pi}$. Using $\Gamma(\frac14)\Gamma(\frac34) = \pi\sqrt{2}$, $\Gamma(\frac54) = \frac14\Gamma(\frac14)$ and $\Gamma(\frac12) = \sqrt{\pi}$ [OLBC10, equations (5.5.3), (5.5.1), (5.4.6)], the limits of H, H', H'' as $a \to 0^+$ are $h_S(0)$, $h_S'(0) = -\pi$ and $h_S''(0)$.

Step 4: uniqueness. The function $W = H - h_S$ is smooth on $(0,\infty)$, satisfies $W''' = -a^2 W' - aW$, and $W, W', W'' \to 0$ as $a \to 0^+$. Fix $a_1 > 0$. For $0 < a \le a_1$, the energy $E = W^2 + W'^2 + W''^2$ satisfies $E' \le (2 + a_1^2 + a_1)E$, by $2|uv| \le u^2 + v^2$. Hence $\mathrm{e}^{-(2+a_1^2+a_1)a}E(a)$ is nonincreasing on $(0, a_1]$ and tends to 0 as $a \to 0^+$, so $E = 0$ there. Thus $H = h_S = h$ on $(0,\infty)$. ∎

The identity of Lemma 5.1 also follows from Lemma 4.1 and the Dixon–Ferrar formula [GR15, entry 6.518]: substitute $s = \sinh t$ in $\Phi_{0,0}(1;y)$, use $K_{1/2}(z) = \sqrt{\pi/(2z)}\,\mathrm{e}^{-z}$, take $\nu = \frac14$ and $x = a^2/4$ in the formula, and use $Y_{1/4} = J_{1/4} - \sqrt{2}\, J_{-1/4}$.

6. Identification with the quartic recurrence

Define

$$V_n \;=\; \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n}, \qquad v_n \;=\; \frac{V_n}{a} \qquad (n \;\geq\; 0), \tag{6.1}$$

and $V_j = v_j = 0$ for $j < 0$. By Corollary 4.5, $v_n > 0$ for every $n \geq 0$. We derive (1.2), specializing [FH25, Lemma 3.4 and equations (3.1)–(3.3)].

If g has C_4-weight ℓ, then

$$\sigma(zg) \;=\; -\tau(\ell) z\, \sigma g. \tag{6.2}$$

For $\ell = 0, 1, 2$ this follows by comparing the twist factors in weights ℓ and $\ell + 1$; for $\ell = 3$ the weight returns to 0, and $\eta^4 = -1$ supplies the sign $\tau(3) = -1$.

Let $L = \partial_z + \mathrm{i}z^3$, where $\partial_z = \frac{1}{2}(\partial_{\operatorname{Re} z} - \mathrm{i}\partial_{\operatorname{Im} z})$. Since $\partial_z \overline{\sigma g(z)} = 0$ and $\partial_z \rho_a(z) = (-a\bar{z} + \mathrm{i}z^3)\rho_a(z)$, integration by parts in (2.2), with no boundary terms because of the Gaussian decay, and (6.2) give, for g of weight ℓ,

$$(Lf, g)_a \;=\; a \int_{\mathbb{C}} f\, \bar{z}\, \overline{\sigma g}\, \rho_a \, \mathrm{d}A \;=\; -\tau(\ell) a (f, zg)_a. \tag{6.3}$$

The polynomials zP_n and LP_n have weights $n+1$ and $n-1$ modulo 4, degrees $n+1$ and $n+3$, and leading coefficients 1 and i. If $j \leq n-7$ and $j \equiv n+1 \pmod 4$, then $\deg LP_j < n$, so (6.3) gives $0 = (LP_j, P_n)_a = -\tau(n) a (P_j, zP_n)_a$. If $j \leq n-5$ and $j \equiv n-1 \pmod 4$, then $\deg zP_j < n$, so $(LP_n, P_j)_a = -\tau(j) a (P_n, zP_j)_a = 0$. Hence

$$zP_n \;=\; P_{n+1} + c_n P_{n-3}, \qquad LP_n \;=\; \mathrm{i}P_{n+3} + d_n P_{n-1}, \tag{6.4}$$

where terms with negative indices are omitted; equivalently, $c_0 = c_1 = c_2 = 0$, and c_n for $n \geq 3$ and d_n for $n \geq 1$ are the coefficients displayed. Among any four consecutive integers exactly one is congruent to 3 modulo 4, so (2.6) gives

$$\varrho(n) \;=\; \tau(n)\varrho(n-3) \quad (n \;\geq\; 3), \qquad \varrho(n) \;=\; -\tau(n-1)\varrho(n-1) \quad (n \;\geq\; 1).$$

Pairing (6.4) with P_{n-3} (for $n \geq 3$) and P_{n-1} (for $n \geq 1$), and using (6.3) and Hermitian symmetry, we obtain

$$\begin{aligned}
\mathrm{i}\varrho(n)\mathfrak{h}_n \;&=\; (LP_{n-3}, P_n)_a \;=\; -\tau(n) a (P_{n-3}, zP_n)_a, \\
c_n \;&=\; \frac{(zP_n, P_{n-3})_a}{\varrho(n-3)\mathfrak{h}_{n-3}} \;=\; \frac{\mathrm{i}}{a}\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-3}}, \\
d_n \;&=\; \frac{(LP_n, P_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} \;=\; -\tau(n-1) a \frac{(P_n, zP_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} \;=\; a\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}};
\end{aligned}$$

the first line holds because the leading term of LP_{n-3} is $\mathrm{i}P_n$, and the last because that of zP_{n-1} is P_n. Thus $c_m = \frac{\mathrm{i}}{a} V_{m-3} V_{m-2} V_{m-1}$ for every $m \geq 0$, including $c_m = 0$ for $m < 3$.

Applying the first relation in (6.4) three times shows that the coefficient of P_{n-1} in $z^3 P_n$ is

$$c_n + c_{n+1} + c_{n+2} \;=\; \frac{\mathrm{i}}{a} V_{n-1} \left(V_{n-2}V_{n-3} + V_n V_{n-2} + V_{n+1} V_n\right).$$

Let $n \geq 1$. The coefficient of P_{n-1} in $\partial_z P_n$ is n, since $\partial_z P_n = nz^{n-1} + \cdots$ lies in the weight space of P_{n-1}. Comparing coefficients of P_{n-1} in $LP_n = \partial_z P_n + \mathrm{i}z^3 P_n$ and using $d_n = aV_{n-1}$, we obtain, with $m = n-1 \geq 0$,

$$V_m \left[1 + \frac{1}{a^2}\left(V_{m-1}V_{m-2} + V_{m+1}V_{m-1} + V_{m+2}V_{m+1}\right)\right] = \frac{m+1}{a}. \tag{6.5}$$

This is the case $d = 4$ of (1.1), and with $V_m = av_m$ it is (1.2).

Finally, $P_0 = 1$, $P_1 = z$, and $P_2 = z^2$. By Lemma 5.1, $h(a) = \int_{\mathbb{C}} \rho_a \, \mathrm{d}A$, and differentiation under this absolutely convergent integral gives

$$h'(a) = -\int_{\mathbb{C}} |z|^2 \rho_a(z) \, \mathrm{d}A(z), \qquad h''(a) = \int_{\mathbb{C}} |z|^4 \rho_a(z) \, \mathrm{d}A(z).$$

With (2.3) and $\varrho(0) = -\varrho(1) = \varrho(2) = 1$, this gives

$$\mathfrak{h}_0 = h, \qquad \mathfrak{h}_1 = -h', \qquad \mathfrak{h}_2 = h''.$$

Hence $h(a) > 0$ and $h'(a) < 0$, so (1.3) is well defined, and (6.1) reproduces it. Thus (v_n) extends the initial values and satisfies (1.2) for every $n \geq 0$. If $(\widetilde{v}_n)$ is another real sequence with these properties and $\widetilde{v}_j = v_j$ for $j \leq m+1$, then (1.2) at index m is linear in $\widetilde{v}_{m+2}$ with coefficient $\widetilde{v}_m \widetilde{v}_{m+1} = v_m v_{m+1} > 0$, so $\widetilde{v}_{m+2} = v_{m+2}$. Induction proves uniqueness. This proves Theorem 1.1: part (i) is Corollary 4.5, and part (ii) has just been shown.

7. Uniqueness of the positive solution

Proof of Theorem 1.2. Fix $\epsilon > 0$ and let (v_n) and (v'_n) be positive solutions of (1.2). Put $T_n = v_n v_{n+1} v_{n+2}$ and $T_{-1} = T_{-2} = 0$, and similarly T'_n. Since $v_n(v_{n-1}v_{n-2} + v_{n+1}v_{n-1} + v_{n+2}v_{n+1}) = T_{n-2} + T_{n-1} + T_n$, the recurrence reads

$$v_n + T_n + T_{n-1} + T_{n-2} = (n+1)\epsilon, \qquad n \geq 0. \tag{7.1}$$

Step 1: a priori bounds. All terms in (7.1) are nonnegative, so $v_n \leq (n+1)\epsilon$ and $T_n \leq (n+1)\epsilon$. Hence, bounding each factor of the bracket in (1.2) by these estimates (for $n = 0, 1$ the products containing a negative index vanish), the bracket is at most $1 + \epsilon^2[n(n-1) + n(n+2) + (n+2)(n+3)] = 1 + 3\epsilon^2((n+1)^2 + 1) \leq K(n+1)^2$ with $K = 1 + 6\epsilon^2$, and therefore

$$v_n \geq \frac{c}{n+1}, \qquad c = \frac{\epsilon}{K}. \tag{7.2}$$

The same bounds hold for v'. With $\delta_n = \log(v'_n/v_n)$, it follows that $|\delta_n| \leq \log K + 2\log(n+1)$.

Step 2: exact linearization. For $x, y > 0$, the logarithmic mean is $L(x,y) = (y-x)/(\log y - \log x)$ if $x \neq y$ and $L(x,x) = x$; equivalently $L(x,y) = \int_0^1 x^{1-t} y^t \, \mathrm{d}t$. Let $a_n = L(v_n, v'_n)$ and $b_n = L(T_n, T'_n)$, so that $v'_n - v_n = a_n \delta_n$ and $T'_n - T_n = b_n D_n$ with $D_n = \delta_n + \delta_{n+1} + \delta_{n+2}$. A logarithmic mean lies between its arguments, so by Step 1

$$a_n \geq \frac{c}{n+1}, \qquad 0 < b_n \leq (n+1)\epsilon. \tag{7.3}$$

Subtracting (7.1) for the two solutions gives

$$a_n\delta_n + b_nD_n + b_{n-1}D_{n-1} + b_{n-2}D_{n-2} \;=\; 0, \qquad n \;\geq\; 0, \tag{7.4}$$

where terms with negative indices are omitted.

Step 3: a localized energy identity. Let $(h_n)_{n\geq 0}$ be real with finite support. Multiplying (7.4) by $h_n^2\delta_n$, summing over n, and collecting the terms with the same b_m gives

$$\sum_{n\geq 0} a_n h_n^2\delta_n^2 + \sum_{m\geq 0} b_m \Big(\sum_{r=0}^{2}\delta_{m+r}\Big)\Big(\sum_{r=0}^{2} h_{m+r}^2\delta_{m+r}\Big) \;=\; 0.$$

For real z_0, z_1, z_2 and h_0, h_1, h_2,

$$\Big(\sum_r z_r\Big)\Big(\sum_r h_r^2 z_r\Big) \;=\; \Big(\sum_r h_r z_r\Big)^2 + \sum_{0\leq r<s\leq 2}(h_r - h_s)^2 z_r z_s.$$

Dropping the nonnegative squares, using $2|z_rz_s| \leq z_r^2 + z_s^2$ and (7.3), we obtain

$$c\sum_{n\geq 0}\frac{h_n^2\delta_n^2}{n+1} \;\leq\; \frac{\epsilon}{2}\sum_{m\geq 0}(m+1)\sum_{0\leq r<s\leq 2}(h_{m+r}-h_{m+s})^2\big(\delta_{m+r}^2+\delta_{m+s}^2\big). \tag{7.5}$$

Step 4: a power weight. Let $0 < p < 1$, put $w_n = (n+1)^{-p}$, and

$$S_p \;=\; \sum_{n\geq 0}\frac{\delta_n^2}{(n+1)^{1+2p}},$$

which is finite by the logarithmic bound of Step 1. For an integer $R \geq 1$, the sequence $h_n = (w_n - w_R)_+$ is supported on $n < R$, and $|h_i - h_j| \leq |w_i - w_j|$. By the mean value theorem, $|w_{m+r} - w_{m+s}| \leq p(s-r)(m+1)^{-p-1}$. In (7.5) the three pairs (r,s) contribute the squared distances 1, 4, 1, so the right-hand side is at most

$$\frac{\epsilon p^2}{2}\sum_{m\geq 0}\frac{5\delta_m^2 + 2\delta_{m+1}^2 + 5\delta_{m+2}^2}{(m+1)^{1+2p}} \;\leq\; \frac{\epsilon p^2}{2}\cdot 12\cdot 27\, S_p \;=\; 162\,\epsilon p^2 S_p,$$

where we used $(m+r+1)/(m+1) \leq 3$ and $1+2p \leq 3$. As $R\to\infty$, $h_n \uparrow w_n$, and monotone convergence on the left of (7.5) gives $cS_p \leq 162\,\epsilon p^2 S_p$. Choosing $p = 1/(18\sqrt{K})$ makes $162\,\epsilon p^2 = c/2$, so $cS_p \leq \frac{c}{2}S_p$. Since $S_p < \infty$, $S_p = 0$, hence $\delta_n = 0$ and $v_n' = v_n$ for every n. ∎

With a_n, b_n held fixed, (7.4) is the Euler–Lagrange equation of the quadratic form $\sum_n a_n\delta_n^2 + \sum_n b_n(\delta_n + \delta_{n+1} + \delta_{n+2})^2$. In the variables $u_n = \log v_n$, (7.1) itself is the coordinatewise stationarity condition of the formal energy $\sum_n\big[\mathrm{e}^{u_n} + \mathrm{e}^{u_n+u_{n+1}+u_{n+2}} - (n+1)\epsilon\, u_n\big]$, whose increments along finitely supported variations are well defined and strictly convex. Convexity alone does not compare two infinite sequences; the power-weighted estimate of Steps 3 and 4 controls the boundary at infinity.

AI-use disclosure

The proof of Theorem 1.1 was developed with assistance from OpenAI Codex using GPT-5.6 Sol (`gpt-5.6-sol`) at high and extra-high reasoning effort. The manuscript was revised, and the Lean formalization of Theorems 1.1 and 1.2 was produced, with assistance from Anthropic's Claude Code using Claude Opus 5.5, Claude Fable 5 and Claude Haiku 5.5. The weighted energy estimate in the proof of Theorem 1.2 was obtained in a consultation with OpenAI's GPT-6 Pro, which was also used for independent referee-style checks of the proofs, the citations and the Lean statements. These systems were used as interactive research assistants to develop and stress-test the arguments, audit citations and calculations, write the formal proofs, and revise the manuscript. The author directed the research, selected and checked the final arguments, and takes responsibility for all mathematical claims, citations, computations, and wording.

Code availability

The Lean 4 formalization of Theorems 1.1 and 1.2 is in the directory `lean` of release `v1.1` of the repository github.com/skypher/painleve (Lean 4.33.0, Mathlib commit `db584cd6d46c`). The two statements, `Painleve.Stmt_main` and `Painleve.Stmt_unique`, are proved by `Painleve.main_theorem` and `Painleve.unique_theorem`. These proofs use only the axioms `propext`, `Classical.choice` and `Quot.sound`; the repository README gives the build commands and the file `lean/Check.lean` that prints both checks. The directory `runs/uniq` contains the numerical scans of initial values that were used as a sanity check for Theorem 1.2.

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